REFERENCE SET · THIRTEEN PLATES · OCTOBER 2026

AI operating system

Thirteen diagrams, each showing a different behaviour of the same system. Every figure was measured against the factory's code on 18 October 2026, and every plate was drawn from the measurement. Prepared as reference artwork for the public `ai-systems` repository.

13	0	0	October 2026
plates	figures unverified	figures contested	compiled

DERIVED FROM

`the software factory` — *private; figures at 87fae286 (18 October), citations at 5739a97 (14 September)*

`four product repositories` — *read-only, each at a named commit*

`Vercel production deployments` — *product status, 14 September 2026*

`plans/ais-plate-claims.json` — *every citation resolved by a script*

Every figure in this set carries its own evidence tier. There is no blanket disclaimer, because a single global caveat tells a reader that something is unverified without telling them which thing, and nothing on any page can then be trusted. The tiers cost more to maintain and are worth it.

Visual grammar

The plates form one system. Colour carries meaning and each colour means exactly one thing; form carries meaning independently of colour. Where a distinction matters, it is drawn as topology rather than hue, so it survives a monochrome print.

COLOUR

Teal	Platform substrate: layers, stages, checks, lanes
Purple	Products, and nothing else
Coral	Enforcement: gates, halts, drift-prone surfaces
Gray	Structural: frames, origins, inputs, outputs

FORM

Dashed outline	A gate or boundary condition. Never decorative.
Unit ticks	A count of agents, stages, rounds or attempts. Ticks reconcile by construction.
Numeral	A threshold. Drawn as a number so it cannot be mistaken for a reconciling count.
Direction	On plate 05, outcome. Up and left returns, down terminates, straight continues.
Taper	Distillation. Narrower layers sit above wider ones.
Broken rhythm	A deliberate exception. Plate 08's human gate is the case.

Evidence tiers

Each plate states what its claim rests on. Nothing is tiered higher than its weakest input: a plate whose mechanism is shipped but whose count is audited is labelled for both.

shipped A named file implements it. Ask to see the file.

audited Counted or observed at a named commit, or in a named source on a named date. A count is audited, because no file implements a number.

contested Two sources give different values and the conflict is open. Nothing drawn in this set has an unresolved conflicting source.

unverified No traceable source. Nothing unverified is drawn.

CONTENTS

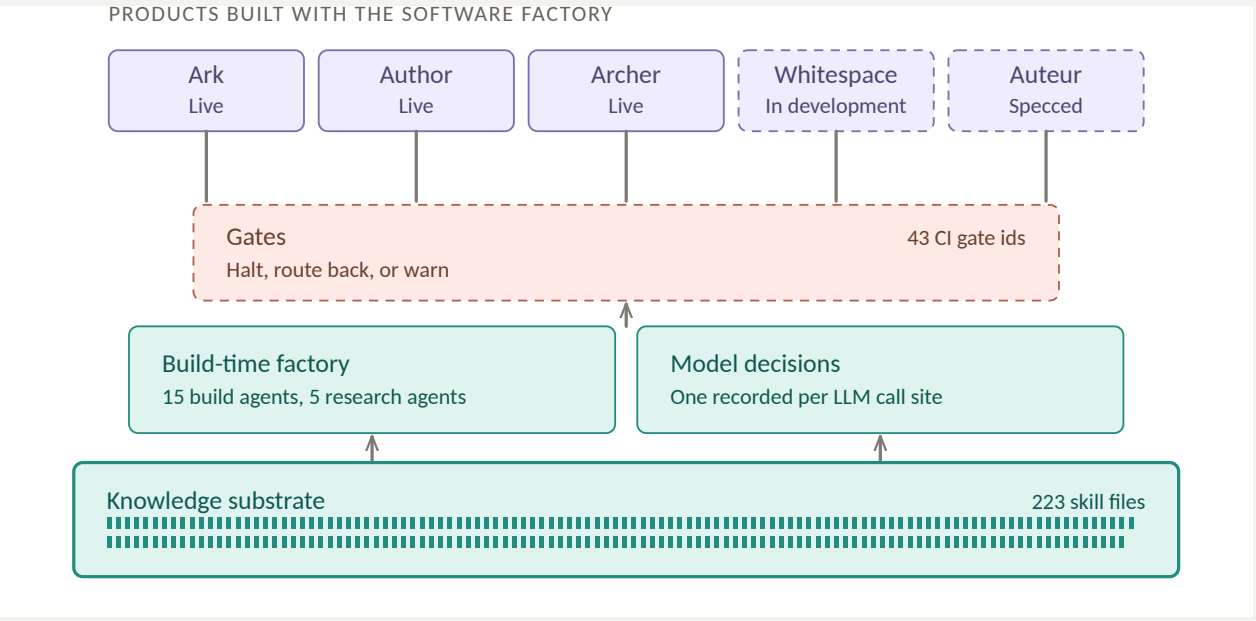
The thirteen plates

Each plate proves one claim. Where two would prove the same thing, one of them is wrong — so the set is not thirteen views of the architecture, it is thirteen separate arguments about it.

01	**The operating system**	A system with a build side and a run side, not a folder of scripts.
02	**The knowledge substrate**	Importance in the corpus is measured, and the measurement is checked.
03	**The compiler spine**	Orchestration with isolation, not a prompt chain.
04	**The contract spine**	One source of truth, and the reach of changing it is computed.
05	**The verification stack**	Four outcomes, and a failure that retrying can fix is told apart from one it cannot.
06	**Context residency**	Budgeting is a measurement, and so is uptake.
07	**The hardening loop**	Promotion requires a proof of convergence.
08	**The learning loops**	Two loops, and in both a person decides what becomes permanent.
09	**The runtime engine**	The packages a product runs on, and the product that runs on them.
10	**Two-lane execution**	The constraint is scoped, not total.
11	**Prompts as specifications**	A prompt is an artifact with a declared shape, graded before it is used.
12	**The writable surface**	Every place a note can land is on a row.
13	**The gate ladder**	Enforcement is a ladder, not a wall.

The operating system

A system with a build side and a run side, not a folder of scripts.



THE TRANSFORMATION

An AI system becomes an organization. Two halves under one corpus, with the same rules governing who may do what on each side. The question it answers is organizational rather than technical: what does each part own, and what must it hand over.

EVIDENCE

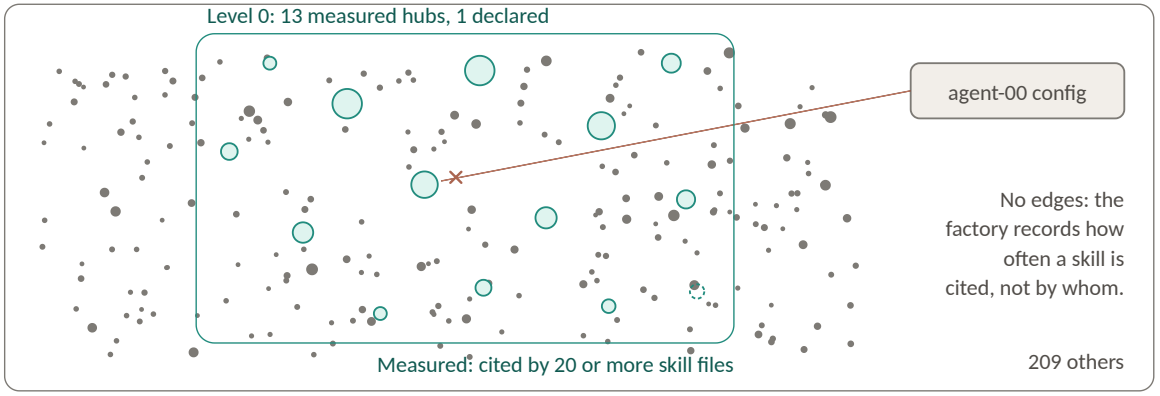
`audited` 223 files, 20 agents, 43 CI gate ids

`audited` product status, 14 Sep

The knowledge substrate

Importance in the corpus is measured, and the measurement is checked.

223 SKILL FILES, EACH DRAWN AT ITS MEASURED CITATION COUNT



ENFORCED IN PRE-COMMIT, AND IN CI AS R33

Three checks on the hubs

The register names each measured hub
Anchors inside all 14 hubs resolve
No live file cites a missing section
Pre-commit; the last two also in R33

The defect behind the third

agent-00 cited agent-constitution §P1-§P2, a section that does not exist, and was handed [section not found]. Fixed 12 Sep.

NINE LAYERS GROUP THE GRAPH; THEY DO NOT RANK IT

Foundational · Governance · Coordination · Engineering · Connectors
Intelligence · Operational · Tracking · Meta

READING

A skill file is a measured hub when twenty or more other skill files cite it, and every node is drawn at the size of its own count. Three files are declared by hand, so the Level-0 set is fourteen, and pre-commit checks hold the hubs' names and anchors.

THE TRANSFORMATION

Knowledge becomes a substrate. Documents stop being reference material an author consults and become an addressable layer the system reads, with importance measured rather than asserted and the measurement itself checked.

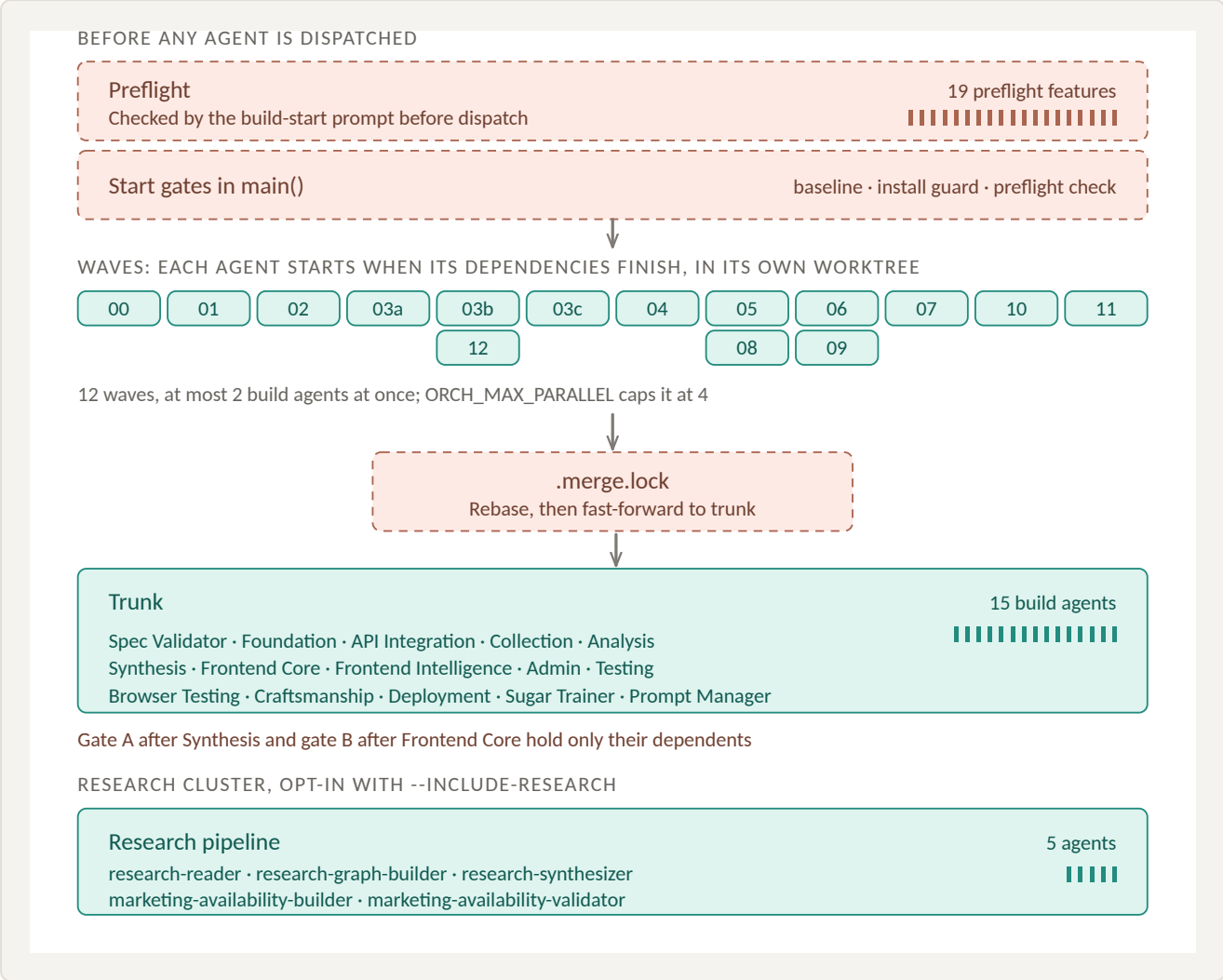
EVIDENCE

```
shipped skill-ref-count.ts, verify-hub-integrity.mjs, audit-hub-citations.mjs
```

```
audited 13 measured hubs, a set of 14
```

The compiler spine

Orchestration with isolation, not a prompt chain.



READING

Every agent whose dependencies have finished starts at once, up to four, each in its own git worktree, and results reach trunk one at a time under a file lock. The strip is the build graph's real twelve waves.

THE TRANSFORMATION

A build process becomes a compiler, and agents become workers. Ambiguous work is decomposed into a dependency graph, scheduled in waves, and executed by workers that cannot see or corrupt each other's state. The isolation is physical, a worktree each, not a convention.

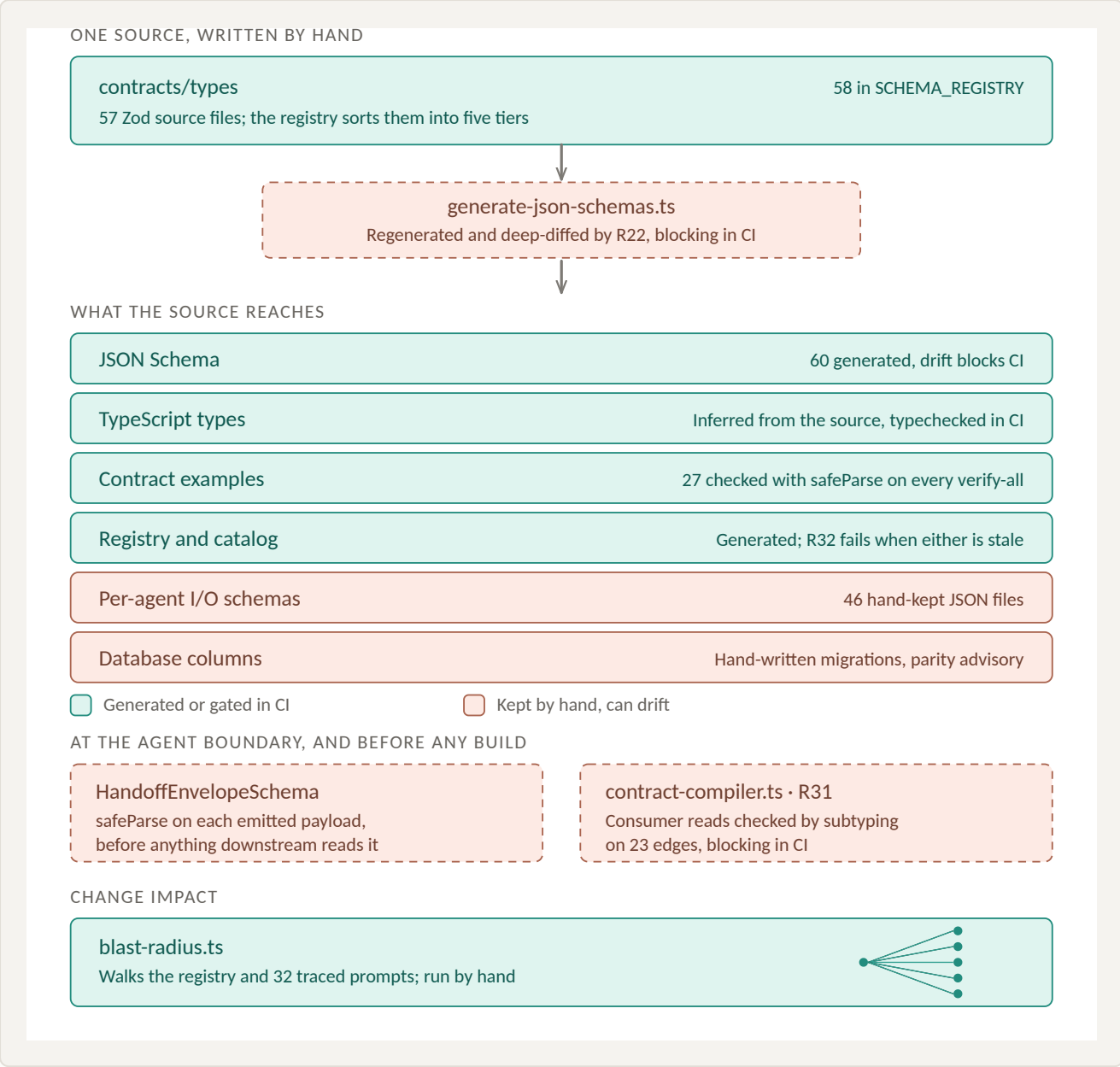
EVIDENCE

shipped agent-scheduler.ts, merge-lock.ts, review-gate.ts

audited 12 waves, at most 2 at once

The contract spine

One source of truth, and the reach of changing it is computed.



READING

Zod is the source. What is generated is gated in CI, and what is kept by hand is drawn in coral because it can drift. Each emitted handoff is parsed against its boundary schema, and the contract compiler checks declared reads by structural subtyping.

THE TRANSFORMATION

Handoffs become contracts. What passes between two agents stops being a message one hopes the next can read, and becomes a typed shape a compiler proves the consumer can accept.

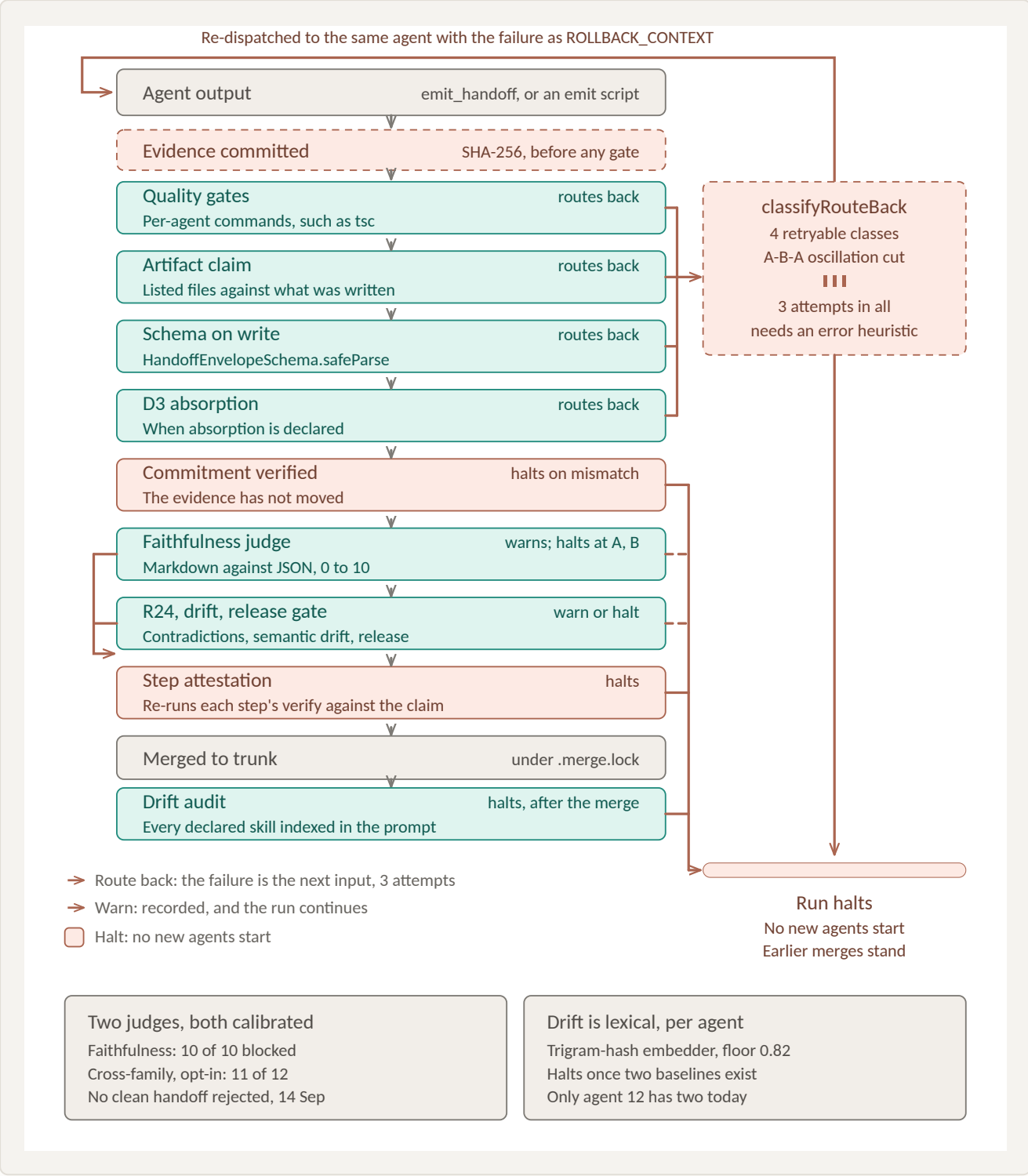
EVIDENCE

`shipped` R22, R31, R32, HandoffEnvelopeSchema

`audited` 58 contracts, 46 hand-kept schemas

The verification stack

Four outcomes, and a failure that retrying can fix is told apart from one it cannot.



READING

Direction encodes outcome: up and to the left returns, down terminates, straight continues. Evidence is committed when an

THE TRANSFORMATION

Agent trust becomes verification. Nothing an agent says about its own work is taken on its word. Each claim is checked against

agent emits. Only the first four checks can route back, and every later failure halts.

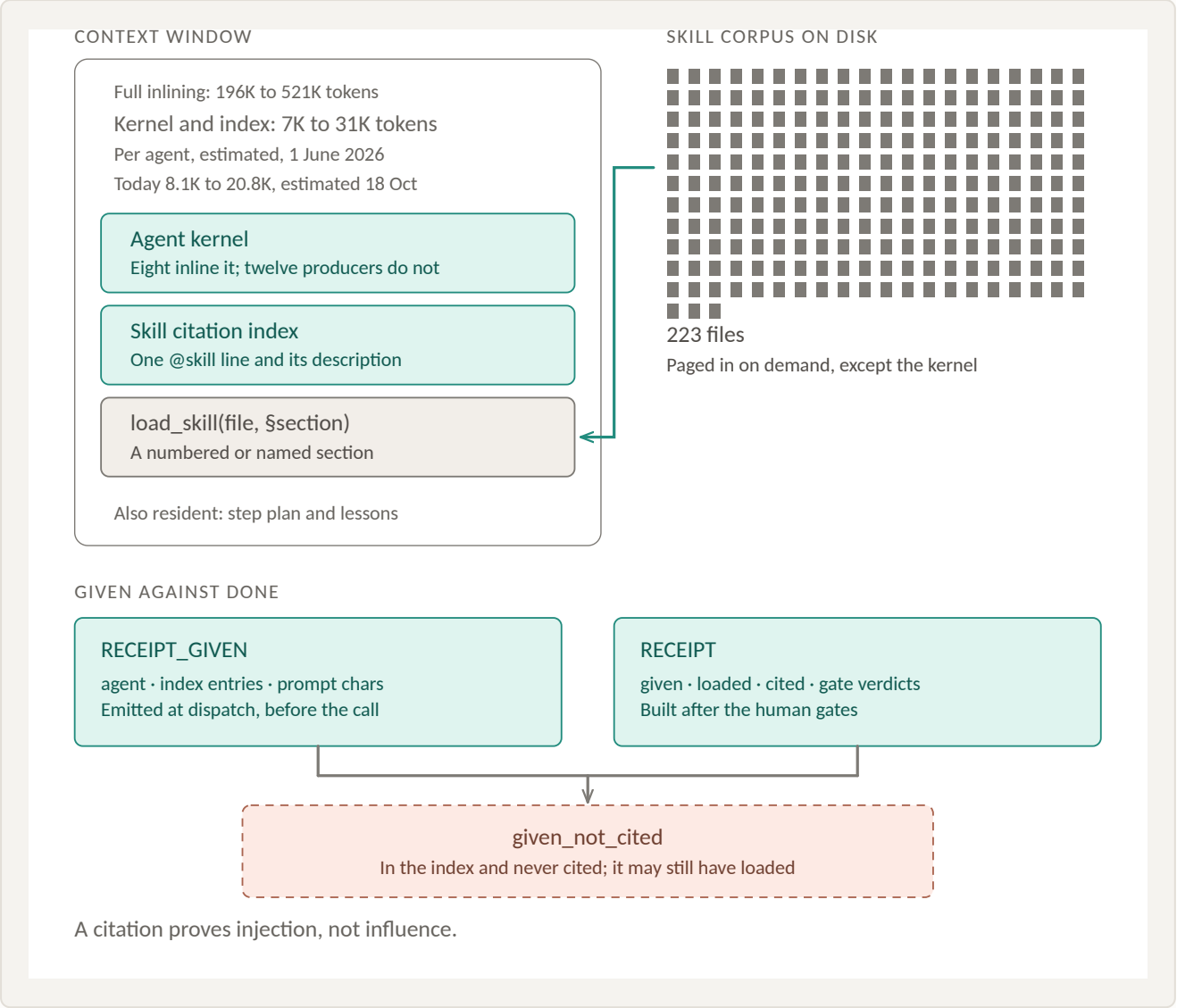
evidence fixed before the question was asked, and a failure that retrying can fix is told apart from one it cannot.

EVIDENCE

```
shipped route-back.ts, evidence-commitment.ts,
semantic-drift.ts, attestation-verifier.ts
```

Context residency

Budgeting is a measurement, and so is uptake.



READING

What sits inside the window and what waits outside it. Token reduction is a cost claim; receipts make the harder one, recording what was given against what the agent cited, and the receipt contract says a citation proves injection, not influence.

THE TRANSFORMATION

Knowledge becomes addressable on demand. An agent carries an index rather than a library, and fetches what the work turns out to need. Residency is budgeted, and uptake is measured separately, because being given a document is not the same as using one.

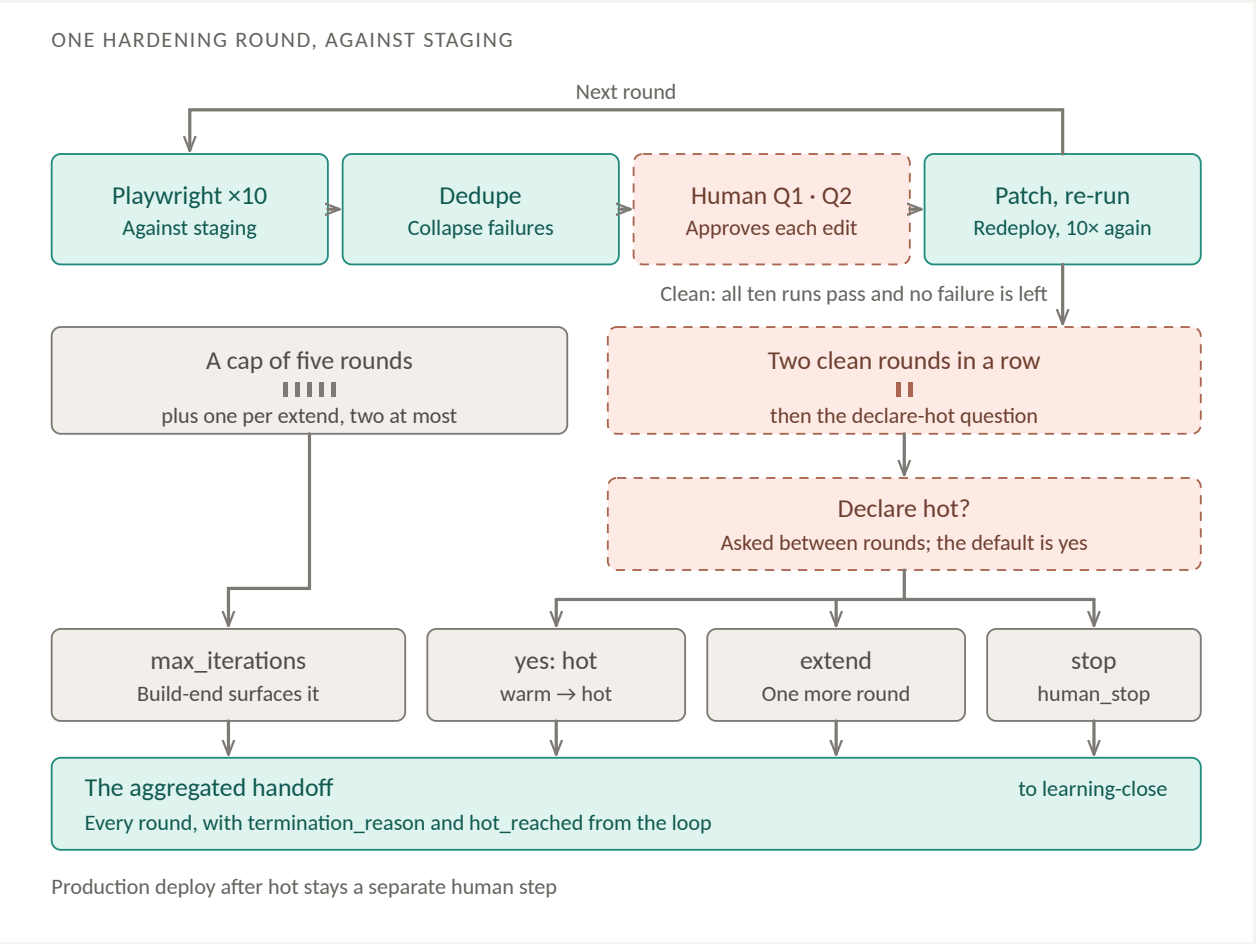
EVIDENCE

`shipped load_skill, skill-sections.ts, receipt.ts`

`audited the reduction and today's loads, estimated`

The hardening loop

Promotion requires a proof of convergence.



READING

Two clean rounds in a row lead to the declare-hot question, which the orchestrator asks between rounds. Yes ends hot, stop ends the loop, extend runs one more round, and the cap moves with each extension.

THE TRANSFORMATION

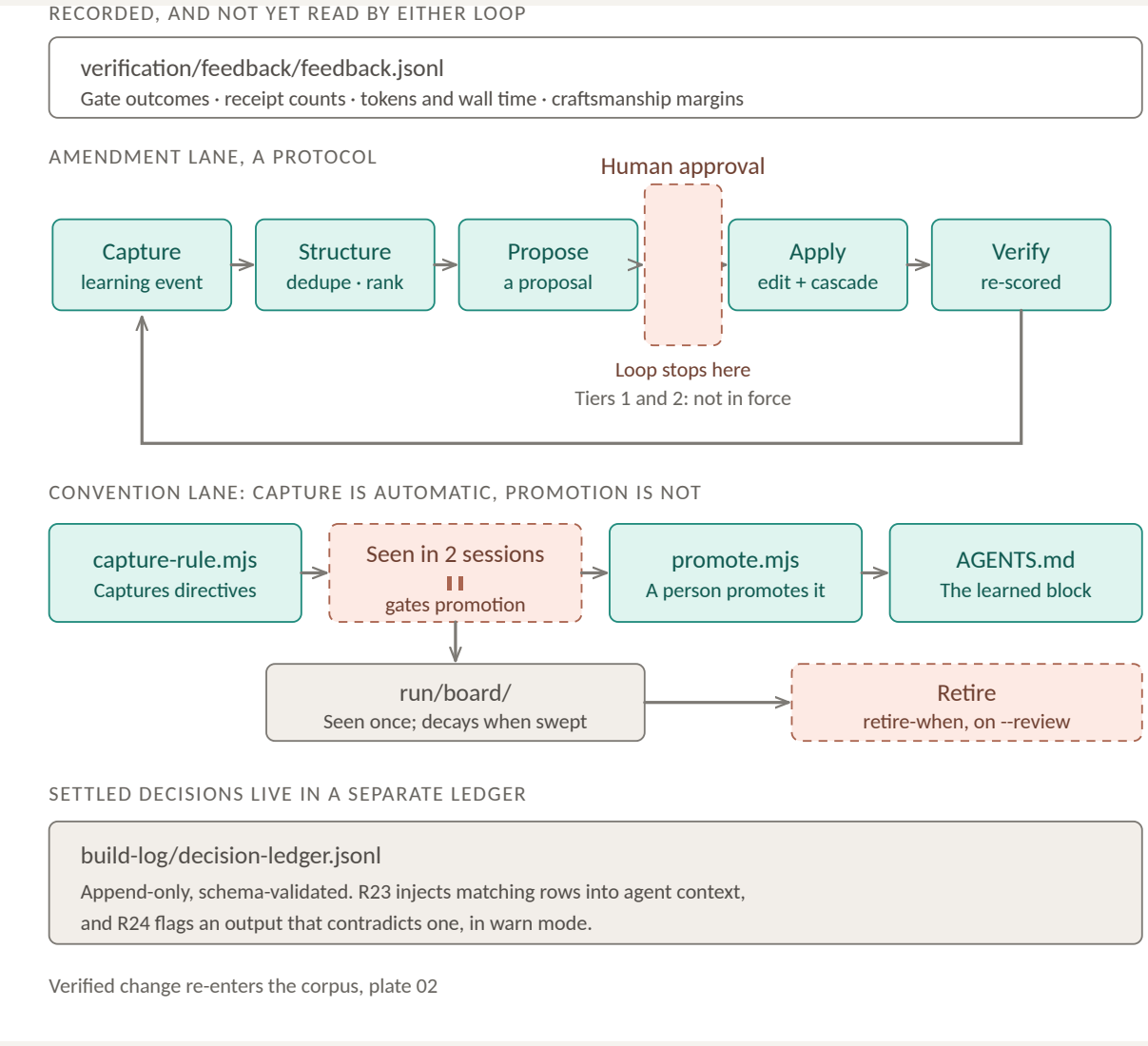
Promotion becomes a proof of convergence. A product is not declared ready because it looks ready. It is declared ready when repeated runs stop producing new failure modes, and a person confirms the ending.

EVIDENCE

```
shipped hardening-state.ts, askDeclareHot, agent-11-steps.md
```

The learning loops

Two loops, and in both a person decides what becomes permanent.



READING

A rule that recurs across two sessions is promoted by a person, and a finding seen once decays on the board. The amendment lane's human approval is the only route, and the decision ledger is a third store the pipeline reads back.

THE TRANSFORMATION

Experience becomes institutional learning. What a run discovers does not stay in that run. Observations become candidate rules, recurrence across sessions promotes them, and a person decides what becomes permanent.

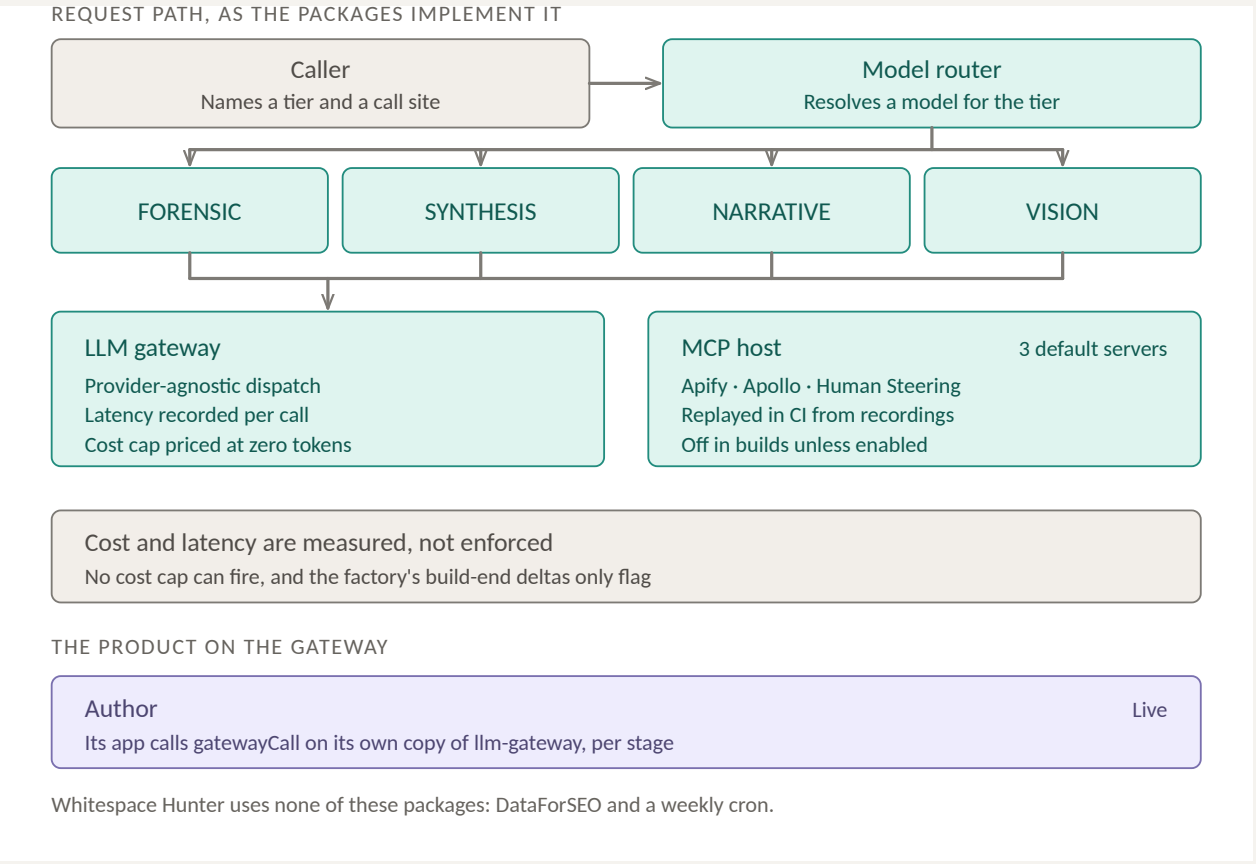
EVIDENCE

shipped capture-rule.mjs, promote.mjs, AGENTS.md, decision-ledger.jsonl

audited the amendment lane, a protocol

The runtime engine

The packages a product runs on, and the product that runs on them.



READING

The caller names a tier, the router resolves a model and the gateway dispatches it. Cost does not stop a run. Author's app is the production caller; Whitespace Hunter, the product with a public URL, uses none of this.

THE TRANSFORMATION

The build's disciplines become a runtime platform. The rules that govern how agents build a product also govern how the finished product talks to a model: one registered call site per decision, one routing layer, one place a failure is named.

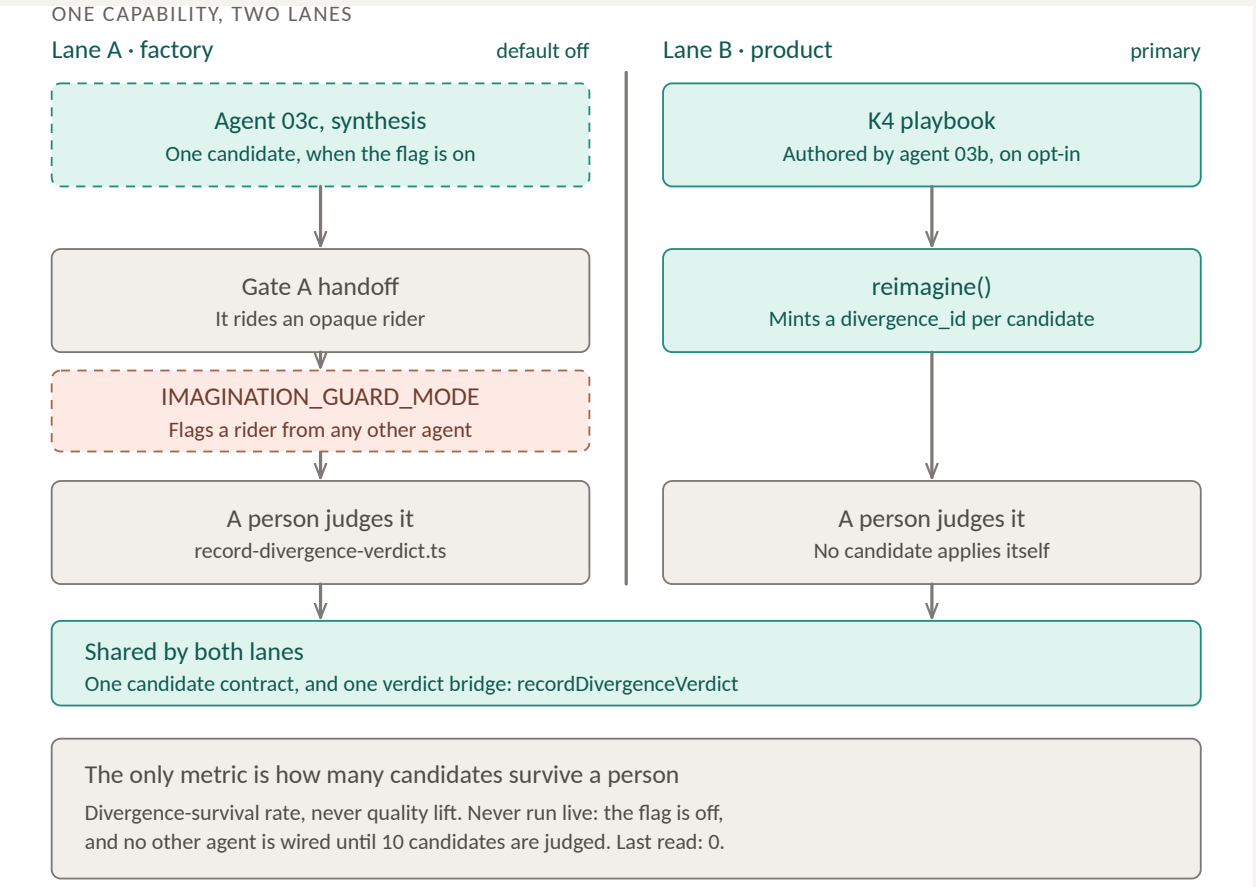
EVIDENCE

`shipped` `llm-gateway, model-router, mcp-infra`

`audited` `Author's gateway client`

Two-lane execution

The constraint is scoped, not total.



READING

Divergence is allowed in two fenced places: one flag-gated candidate from agent 03c at gate A, and an opt-in product runtime. Both share one candidate contract and one verdict path, and a person judges every candidate.

THE TRANSFORMATION

Agent disagreement becomes bounded divergence. A system built entirely for compliance will reproduce known answers. Divergence is allowed in two fenced lanes, judged by a person, and counted only by whether a kept idea survives.

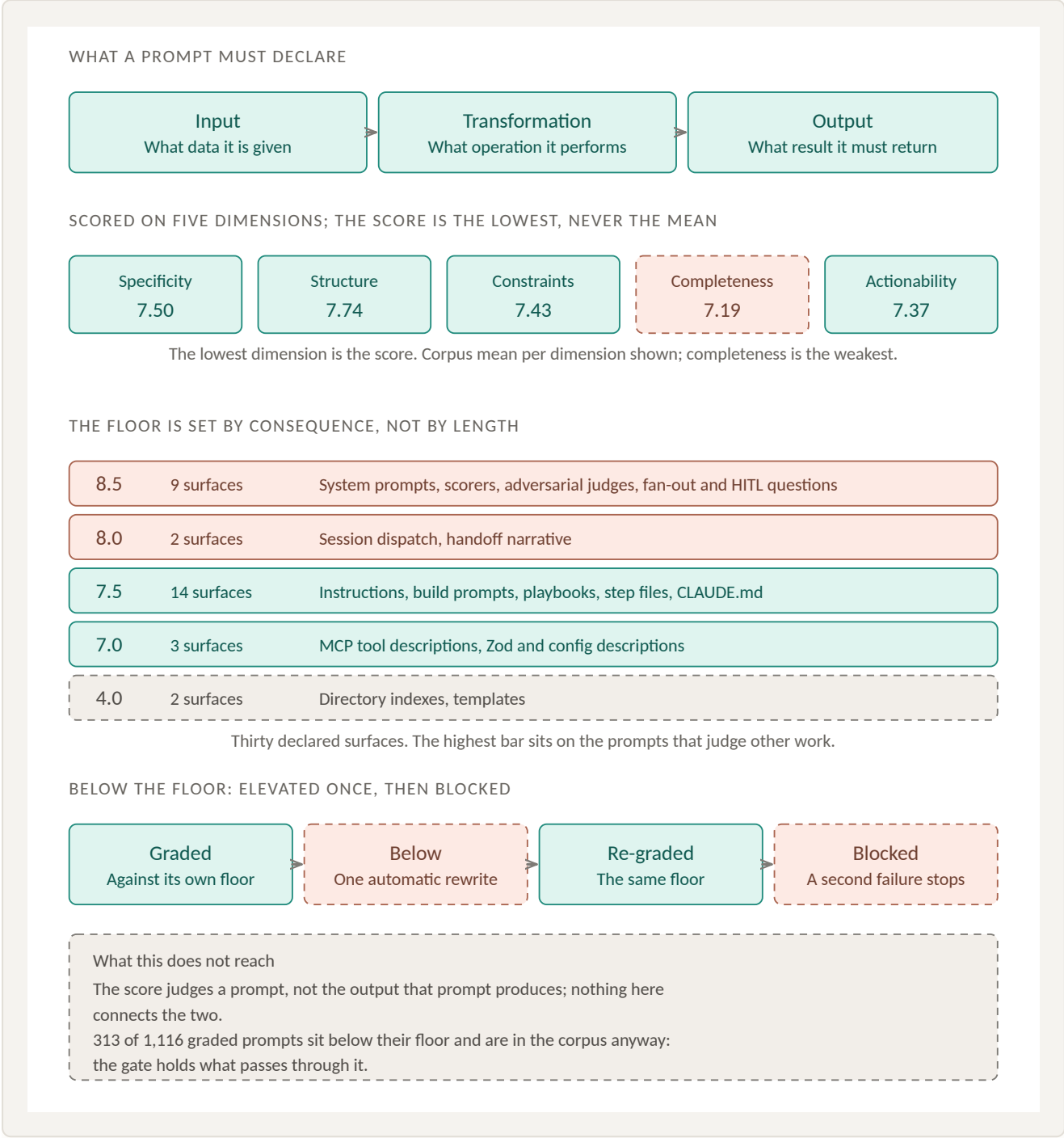
EVIDENCE

```
shipped divergence-candidate.ts,
applyImaginationFlag, validate-handoff.ts
```

```
audited 0 of 10 verdicts; the factory lane is off by
default
```

Prompts as specifications

A prompt is an artifact with a declared shape, graded before it is used.



READING

Three declared specifications, five scoring dimensions of which the LOWEST is the score, and a floor set per surface: 8.5 for the prompts that judge other work, 4.0 for a directory index. Below the floor a prompt is rewritten once, then blocked.

THE TRANSFORMATION

Prompts become specifications. A prompt stops being prose someone wrote well and becomes an artifact with a declared shape, scored against a bar set by what the prompt is for, before it is allowed to run.

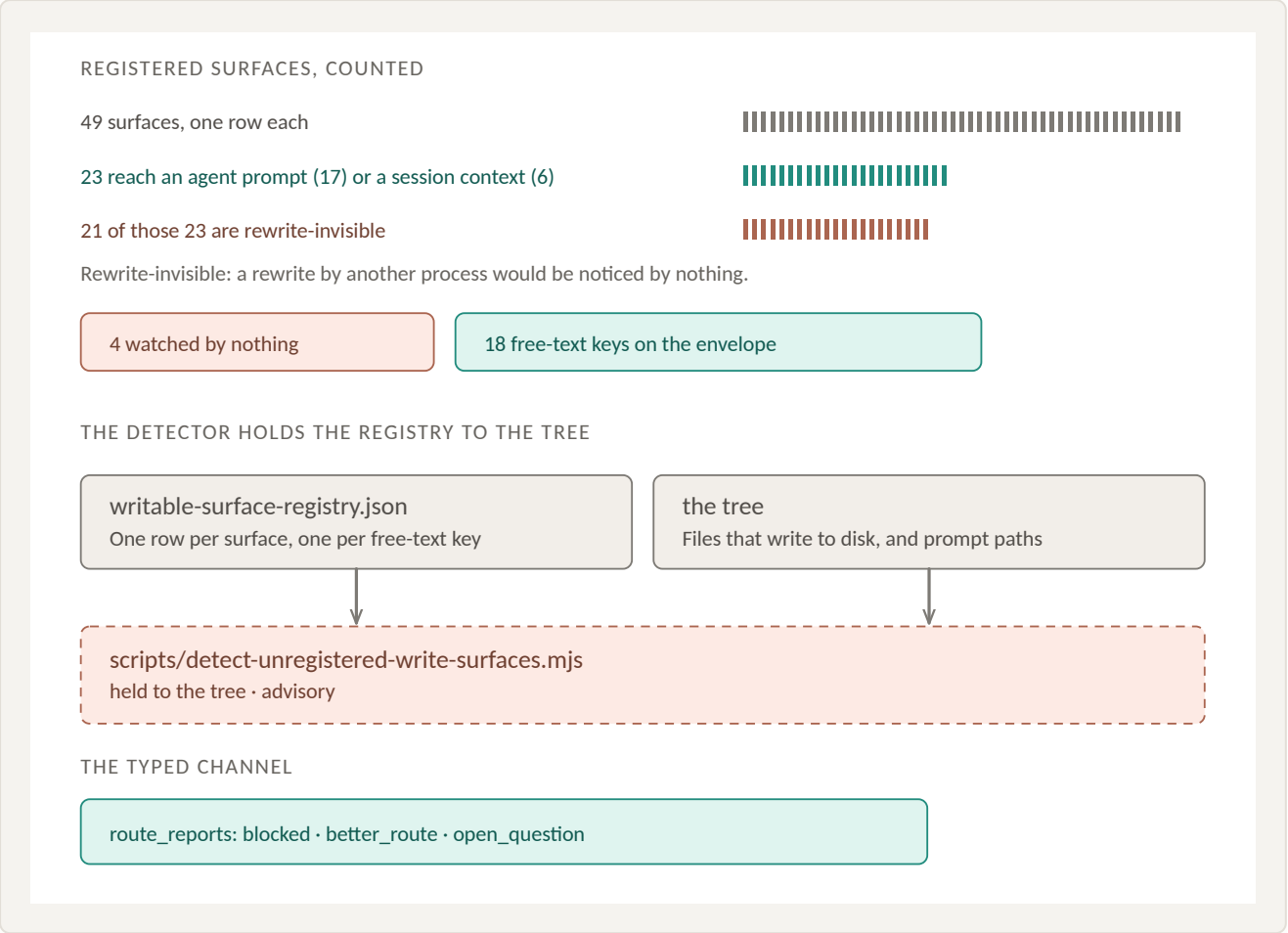
EVIDENCE

shipped craftsmanship-gate.ts, enrich-prompt.ts, SURFACE_FLOORS

audited 30 surfaces, 5 floor levels, 313 of 1,116 below floor

The writable surface

Every place a note can land is on a row.



READING

One row per persistent surface the factory writes: who writes it, who reads it back and where, whether the content is free text, whether a rewrite would be noticed, what watches it. Forty-nine rows; twenty-three reach a prompt or a session; twenty-one of those are rewrite-invisible; four are watched by nothing. A detector holds the registry to the tree.

THE TRANSFORMATION

Pressure makes a channel. An agent with no sanctioned place to say something finds an unsanctioned one, so every channel is inventoried and none is banned.

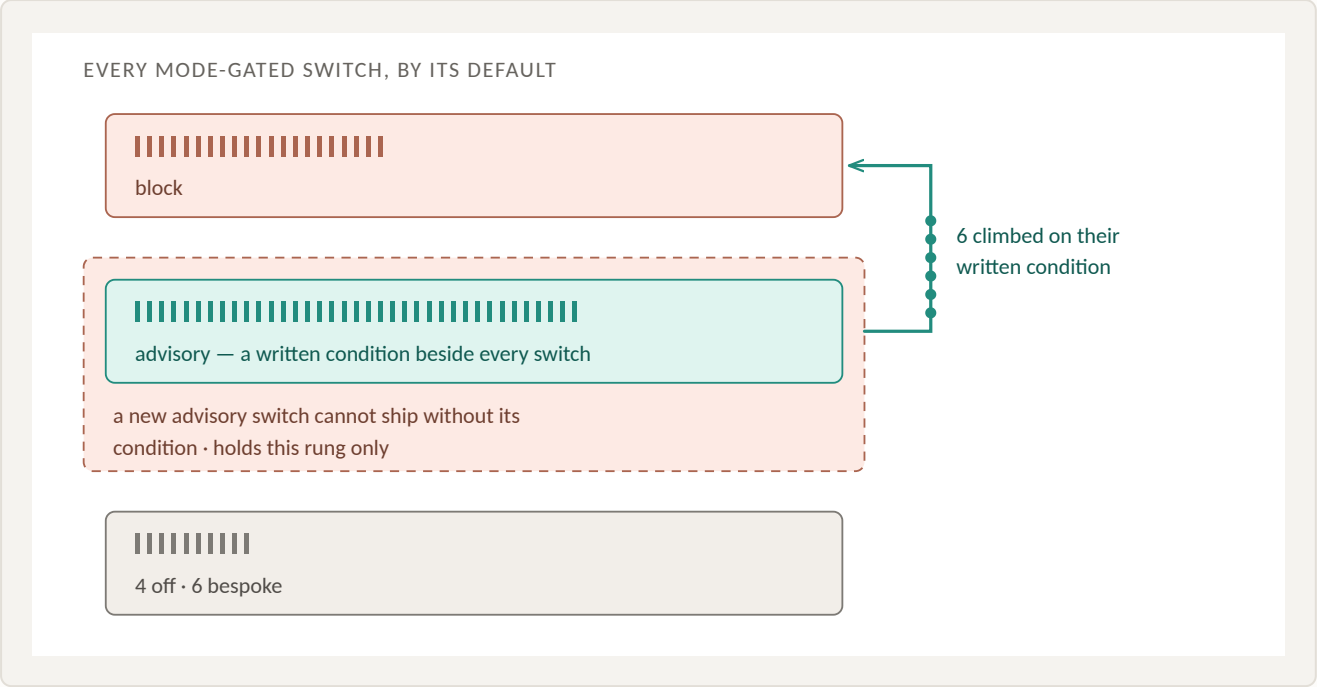
EVIDENCE

shipped writable-surface-registry.json, detect-unregistered-write-surfaces.mjs (advisory)

audited 49 surfaces, 23 reaching, 21 rewrite-invisible, 4 unwatched, 18 free-text keys

The gate ladder

Enforcement is a ladder, not a wall.



READING

Sixty-eight switches: twenty-one default to block, thirty-seven to advisory, four to off, six to a bespoke default. Every advisory switch carries, beside it, the written condition that ends that, and a gate refuses a new advisory switch without one. Six climbed on theirs.

THE TRANSFORMATION

A gate reports pass, fail or unevaluated, and unevaluated is never a pass; a gate that refuses to run exits 2 in every mode.

EVIDENCE

shipped `verify-flip-conditions.mjs, env-mode.ts`

audited 68 switches: 21 block, 37 advisory, 4 off, 6 bespoke; 6 flips on their written condition

Structure

The counts that carry the structural argument, each measured at factory commit 87fae286 on 18 October; the citations behind them were audited at 5739a97. A count is audited, not shipped, because no file implements a number; the audit records how each one was measured.

FIGURE	VALUE	TIER	SOURCE	CHANGED
Skill files	223	audited	skills/, per-layer indexes excluded	was 152, contested
In a manifest, parked	192, 31	audited	agent configs, retired-skills.json	new
Architecture layers	9	audited	architecture_layer in frontmatter	was 8 or 9
Agents	20: 15 build, 5 research	audited	configs/agent-*.config.ts	unchanged
Build waves	12, at most 2 at once	audited	dependsOn, the scheduler's rule	new
CI gate ids	43	audited	verify-all.ts	was 30, contested
Preflight features	19	audited	prompts/feature-descriptor.md	were called gates
Registered contracts	58, in five tiers	audited	SCHEMA_REGISTRY	was 19, contested
Generated JSON Schemas	60	audited	contracts/schemas, outside agents/	was 43, contested
Per-agent I/O schemas	46	audited	contracts/schemas/agents	new
Measured Level-0 hubs	13, in a set of 14	audited	skill-ref-counts.json	was 11
Static prompt load	8,146 to 20,813 tokens	audited	token-budget-baseline.json	new

Mechanism

Every mechanism the set draws and the file that implements it. The amendment lane is the one row that is a protocol rather than code.

MECHANISM	WHAT IT DOES	IMPLEMENTED IN	TIER
Dependency-wave dispatch	Starts each agent once its dependencies finish, up to 4	`agent-scheduler.ts`	shipped
Merge serialisation	Rebase, then fast-forward, under .merge.lock	`merge-lock.ts`	shipped
Schema on write	safeParse on each emitted payload	`orchestrator`	shipped
Contract compiler	Declared reads checked by structural subtyping (R31)	`contract-compiler.ts`	shipped
Change impact	Breadth-first over the registry's imports; run by hand	`blast-radius.ts`	shipped
Evidence commitment	SHA-256 when an agent emits, verified before any judge	`evidence-commitment.ts`	shipped
Route-back	3 classes, a matching heuristic, A-B-A cut, 3 attempts	`route-back.ts (classifyRouteBack)`	shipped
Faithfulness judge	Narrative against JSON; warns, blocks at gates A and B	`verify-handoff-faithfulness.ts`	shipped
Semantic drift	Trigram-hash embedder, floor 0.82, enforcing at two baselines	`semantic-drift.ts`	shipped
Step attestation	Re-runs each step's verify; blocking	`attestation-verifier.ts`	shipped
Drift audit	Every declared skill indexed in the prompt; after the merge	`drift-audit.ts`	shipped
Section loading	A numbered or named section, never a silent whole file	`skill-sections.ts`	shipped
Receipts	RECEIPT_GIVEN at dispatch, RECEIPT after the gates	`receipt.ts`	shipped
Hardening termination	Two clean rounds, then declare hot: yes, extend or stop	`hardening-state.ts, askDeclareHot`	shipped
Convention promotion	Two sessions, or a recorded reason, by a person	`promote.mjs, AGENTS.md`	shipped
Decision ledger	Append-only, schema-validated, injected by R23	`decision-ledger.jsonl`	shipped
Amendment lane	A protocol whose human approval is the only tier in force	`learning-loop.md`	audited
LLM gateway	Provider adapters and per-call latency; its cost cap cannot fire	`llm-gateway`	shipped
MCP host	Three default servers, replayed in CI	`mcp-infra`	shipped
Craftsmanship gate	SHA-256 per prompt file; five dimensions, the lowest is the score; elevated once, then blocked	`craftsmanship-gate.ts, enrich-prompt.ts`	shipped

MECHANISM	WHAT IT DOES	IMPLEMENTED IN	TIER
Surface floors	30 surfaces, five floor levels from 4.0 to 8.5	`craftsmanship-surface-policy.ts`	shipped
Surface registry	39 rows: writer, read-back, free text, tamper evidence, watcher	`writable-surface-registry.json`	shipped
Surface detector	Holds the registry to the tree; advisory	`detect-unregistered-write-surfaces.mjs`	shipped
Route reports	blocked, better_route, open_question; at most 10; locked at emission	`route-report.ts, receipt.ts`	shipped
Mode switches	58, one reading; block, advisory or off	`env-mode.ts`	shipped
Flip ratchet	A new advisory switch cannot ship without its written condition	`verify-flip-conditions.mjs`	shipped
Imagination lanes	One flag-gated candidate at gate A; an opt-in product runtime	`runtime-imagination, validate-handoff.ts`	shipped

Products

Built with the factory. Status is audited from each project's Vercel production deployments on 14 September 2026; the repositories are private.

PRODUCT	WHAT IT IS	STATUS
Ark	A brand book researched and written from a URL	`Live; deployed 7 Sep`
Author	A content playbook compiler, from a brand's URL	`Live; deployed 26 Jul`
Archer	B2B sales intelligence for a target account	`Live; deployed 26 Jul`
Whitespace Hunter	Runway scoring for emerging categories	`In development`
Auteur	AI creative production	`Specced`

Not drawn

FIGURE	PREVIOUS VALUE	WHY IT IS OUT
Residency class counts	~~`12 resident / 60 index / 132 paged`~~	No source, and the counts summed to the wrong total.
Build-time compression	~~`2–4 weeks against 6–9 months`~~	A comparative claim with no measured baseline on either side.
Intervention reduction	~~`up to 80% fewer manual interventions`~~	No instrument produces this number. The phrase 'up to' is doing all the work.
Trend surge magnitudes	~~`18x, +1800%`~~	One run's output from a real tool, not a headline figure.
Type-check saving	~~`about 6 seconds per dispatch`~~	Not measured, and no plate makes a claim it would support.
Graded prompt corpus	~~`471`~~	Not re-counted in this audit.

One defect is recorded here because it is worth publishing rather than despite being embarrassing. `agent-00.config.ts` cited a section anchor that did not exist, so the spec validator — first in the pipeline, the gate every build passes through — was handed the literal string `[section not found]` where its constitution should have been. A citation that silently resolves to an error string is worse than one that throws, because the pipeline keeps running and everything downstream is confidently built on nothing. It was fixed on 12 September, and since 14 September a citation to a section that does not exist fails in pre-commit and in CI.

The plates are compiled, not drawn. A grammar module holds the colour roles, ticks, gates and taper as primitives, and each plate is written in terms of those, so a grammar violation is impossible to express rather than something a reviewer has to catch. It is the argument plate 04 makes about schemas, applied to the artwork. The source and its regeneration instructions are in `tools/diagram-compiler`. The plates are committed as SVG, not raster: they render natively on GitHub, appear in diffs, and let a reviewer watch the architecture change over time.